

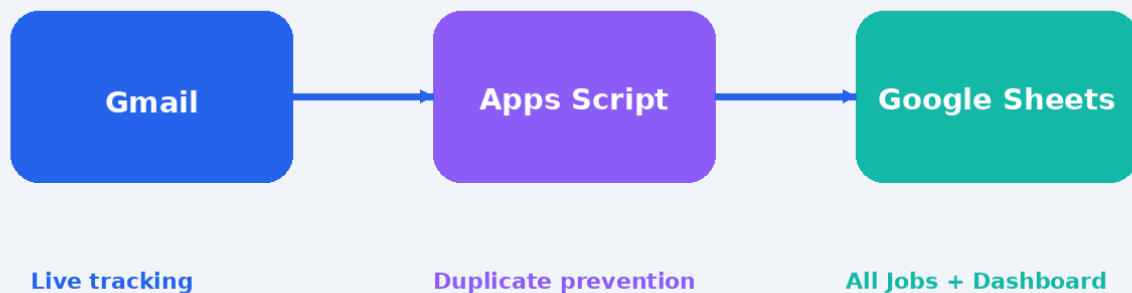
GMAIL JOB TRACKER

Automated Recruiter & Application Tracking

Professional Setup, Architecture & Demo Guide

The solution

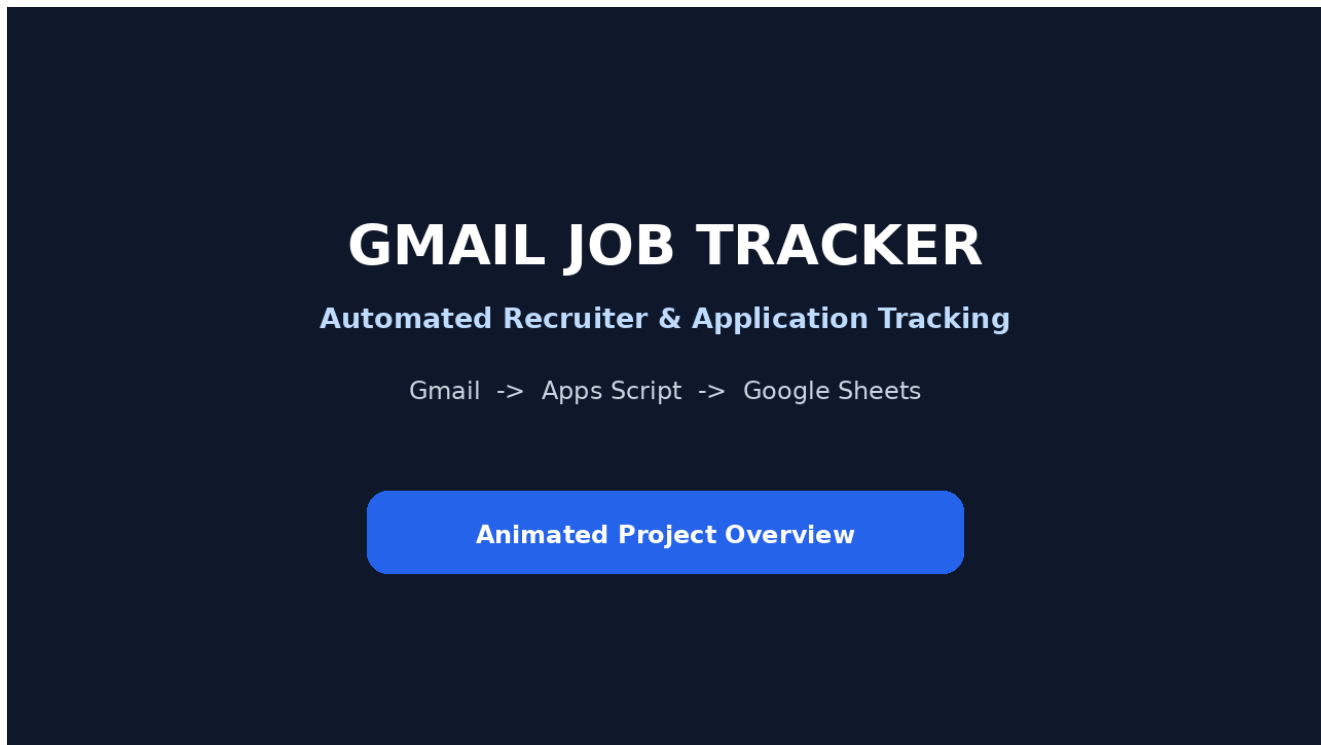
A transparent rule-based automation converts Gmail activity into structured tracking.



Google Apps Script | Gmail | Google Sheets | Automated Dashboard

Privacy-first public edition: use synthetic demo data and never publish real Gmail, recruiter, application, or Google Sheet identifiers.

1. Project Overview



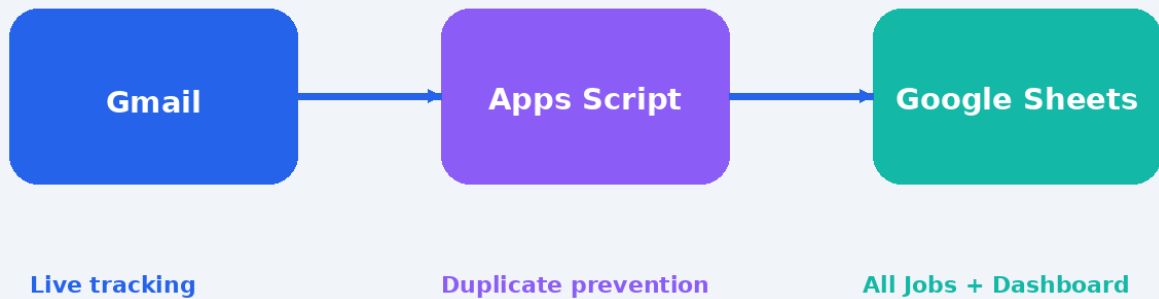
The Gmail Job Tracker converts job-search activity in Gmail into a structured Google Sheets system. It automatically captures recruiter/vendor outreach, direct application confirmations, later status changes, historical activity, and dashboard totals.

- Creates one daily tracker sheet named YYYY-MM-DD from a TEMPLATE tab.
- Tracks recruiter/vendor emails you send and identifies follow-ups.
- Detects direct applications and common ATS/job-board confirmations.
- Updates recruiter and portal statuses from later Gmail messages.
- Maintains a permanent All Jobs master sheet and a hidden _Processed state sheet.
- Supports hourly live tracking plus batched historical Gmail reconstruction.

2. Architecture

The solution

A transparent rule-based automation converts Gmail activity into structured tracking.



Live automation

- A time-driven trigger runs `runJobTrackerAutomation()`.
- Sent Gmail is checked for recruiter/vendor activity.
- Incoming Gmail is checked for applications and status updates.
- Daily sheets, All Jobs, `_Processed`, and Dashboard are updated as appropriate.
- `LockService` reduces overlapping write operations.

Why `_Processed` matters

Each Gmail message can be stored with a workflow type and Gmail Message ID. Repeated scheduled executions can therefore skip work already completed for the same workflow.

3. Build the Google Sheet

Step 1 - Create TEMPLATE

- Rename the first sheet to TEMPLATE.
- Merge A1:H1 and enter RECRUITER / VENDOR EMAILS SENT.
- Row 2 headers: Date, Company, Recruiter Name, Phone Number, Email, Job Title, Status, Email Link.
- Leave rows 3-6 blank.
- Merge A7:G7 and enter PORTAL / DIRECT JOB APPLICATIONS.
- Row 8 headers: Date, Portal, Company, Position, Application #, Status, Email Link.

The automation inserts recruiter rows while preserving exactly four blank rows above the Portal section.

Recruiter section	Date Company Recruiter Phone Email Job Title Status Email Link
Portal section	Date Portal Company Position Application # Status Email Link

4. Configure Google Apps Script

Step 2 - Create a standalone project

Create a standalone Apps Script project named Job Tracker Automation. Add every .gs file from the repository src/ directory. Apps Script loads project files together, so functions can call one another across files.

Step 3 - Configure the spreadsheet ID

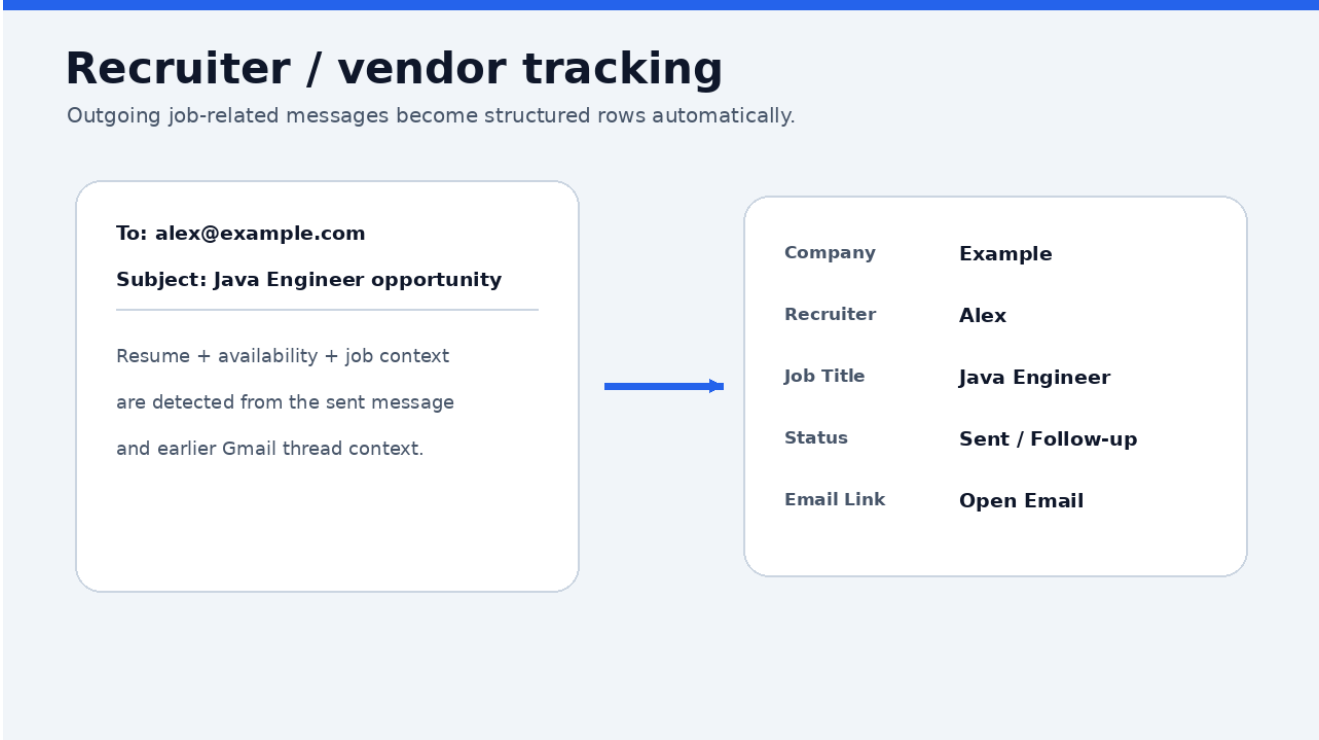
The public source uses: `const SPREADSHEET_ID = 'YOUR_GOOGLE_SHEET_ID_HERE' ;`

Replace the placeholder only in your private Apps Script deployment. Never commit the real Sheet ID back to the public repository.

Step 4 - Authorize and test

- Run `testSpreadsheetConnection` and approve the requested Google permissions.
- Run `createTodaySheet` and confirm a YYYY-MM-DD sheet is copied from TEMPLATE.
- Run `createTodaySheet` again and verify the existing daily sheet is reused.

5. Recruiter & Application Tracking



Recruiter/vendor tracking

- testRecruiterSentEmails evaluates sent-mail job/recruiting language and earlier Gmail thread context.
- syncRecruiterSentEmailsToSheet writes a structured recruiter row, Gmail link, and _Processed entry.
- Records can include company, recruiter name/email, phone, job title, and Sent vs Follow-up Sent.

Portal/direct application tracking

- testPortalApplicationEmails checks incoming mail for strong application-confirmation language.
- syncPortalApplicationsToSheet writes confirmed applications into the portal section.
- Sender mappings include Workday, Greenhouse, Lever, iCIMS, Indeed, LinkedIn, Dice, Monster, ZipRecruiter, Jobvite, Taleo, SmartRecruiters, and others.
- Generic job alerts are down-weighted unless strong application evidence is present.

6. Status Matching & Progression

Status progression

Later Gmail replies update existing records instead of creating duplicates.



Ambiguous matches are rejected instead of guessed.

Recruiter statuses

Sent -> Follow-up Sent -> Replied -> RTR -> Submitted -> Assessment -> Interview -> Offer. Terminal outcomes can include Rejected and Closed.

Portal statuses

Applied -> Assessment -> Interview -> Offer. Terminal outcomes can include Rejected and Withdrawn.

Thread matches receive strong preference. Additional evidence includes recruiter email, company, job title, or application position. Ambiguous matches can be rejected rather than guessed, and status rules prevent noisy regressions.

7. Live Automation

Step 5 - Test the master function

- Run `runJobTrackerAutomation` manually once.
- Verify recruiter sent sync, recruiter status sync, portal application sync, application status sync, and Dashboard refresh complete.

Step 6 - Create the hourly trigger

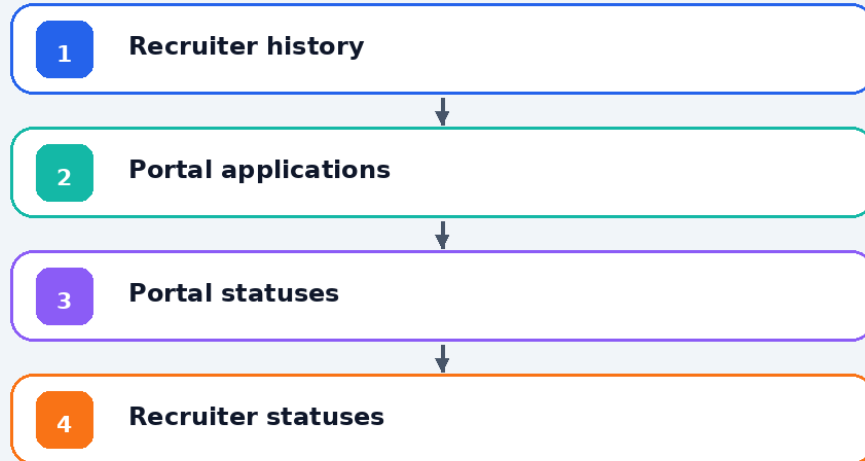
- Open Apps Script > Triggers.
- Click Add Trigger.
- Choose `runJobTrackerAutomation`.
- Event source: Time-driven.
- Choose Hour timer > Every hour.
- Save.

Apps Script time-driven triggers are scheduled automation, not a hard real-time event stream.

8. All Jobs & Historical Backfill

Historical reconstruction

Older Gmail is imported in scheduled batches instead of one long execution.



Temporary triggers + Script Properties + _Processed

Step 7 - Create All Jobs

Run createAllJobsSheet once to create the permanent master tracking tab and status dropdowns.

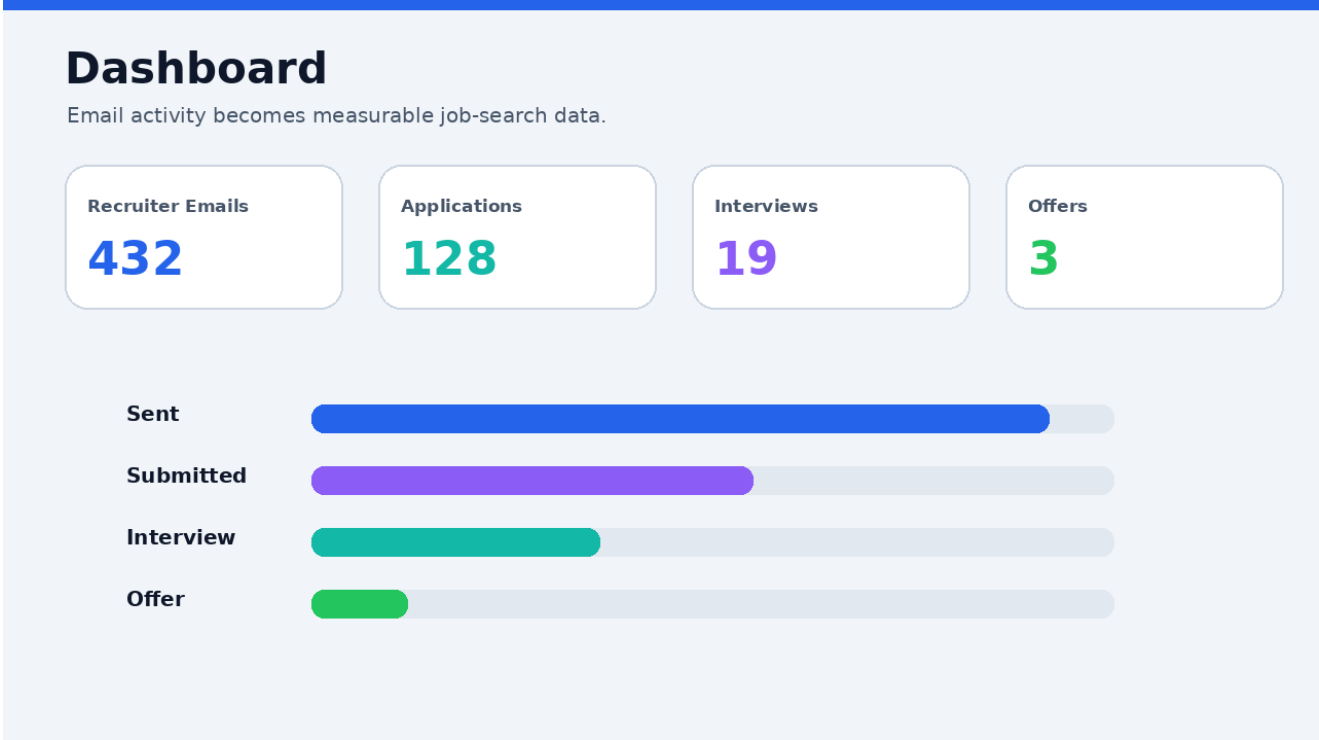
Step 8 - Run historical stages sequentially

- startAllJobsRecruiterBackfill
- startAllJobsPortalBackfill
- startAllJobsPortalStatusBackfill
- startAllJobsRecruiterStatusBackfill

Do not repeatedly rerun a backfill starter while that stage is active. Use checkAllJobsBackfillProgress to inspect progress.

Historical functions use temporary time-driven triggers, Script Properties for saved positions, message-level duplicate tracking, and LockService so a large mailbox can be processed across multiple normal Apps Script executions.

9. Dashboard



Run refreshJobDashboard to rebuild summary metrics from All Jobs. Historical totals continue changing while a backfill is still running.

- Today: recruiter/vendor emails and applications.
- Recruiter summary: Sent, Follow-up, Replied, RTR, Submitted, Assessment, Interview, Rejected, Offer, Closed.
- Portal summary: Applied, Assessment, Interview, Rejected, Offer, Withdrawn.

10. Verification & Security

Privacy-safe public portfolio

The GitHub project contains code and synthetic visuals - not real job-search data.



Real Sheet IDs



OAuth tokens



Recruiter PII



Gmail exports

Use synthetic data in screenshots and demo recordings.

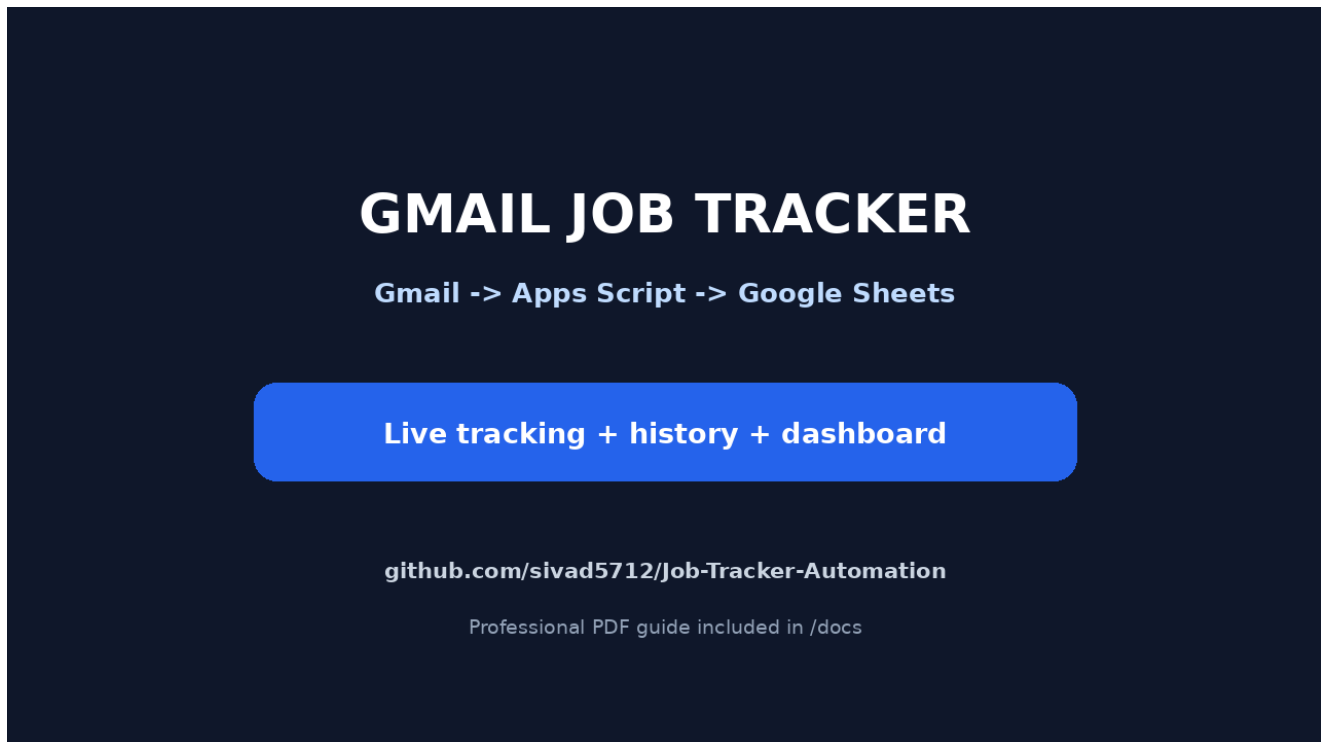
Verification checklist

- TEMPLATE exists and daily YYYY-MM-DD sheets are created.
- Recruiter and portal records appear once per Gmail message.
- Email links open the expected Gmail thread.
- Status updates match the correct record and do not regress unexpectedly.
- All Jobs, Dashboard, and hidden _Processed are present.
- The hourly trigger is visible in Apps Script.

Public GitHub checklist

- Use synthetic recruiter names, companies, emails, phone numbers, job titles, and application numbers.
- Never publish a real Sheet ID, Gmail message/thread ID, OAuth token, API key, or .clasp.json.
- Do not upload screenshots that expose private Gmail or real recruiter/application information.

11. Animated Demo & Portfolio Presentation



The repository includes a captioned 60-second animated explainer designed for GitHub visitors. It communicates the project without exposing a real inbox or recruiter data.

- Problem: job-search information is scattered across Gmail.
- Solution: Apps Script turns email activity into structured tracking.
- Live automation: recruiter messages, applications, statuses, daily sheets, All Jobs, Dashboard.
- Historical backfill: older Gmail is reconstructed in scheduled batches.
- Security: public assets use synthetic data only.

Animated demo: `assets/job-tracker-animated-demo.mp4`

Professional guide: `docs/Gmail_Job_Tracker_Professional_Guide.pdf`

Repository: `github.com/sivad5712/Job-Tracker-Automation`

12. Troubleshooting Quick Reference

Symptom	Likely cause	What to check
Daily sheet not created	TEMPLATE missing/misnamed	Confirm exact TEMPLATE name and rerun createTodaySheet.
Duplicate rows	Processed state missing/reset	Inspect hidden _Processed and workflow type/message ID.
Recruiter email missed	Insufficient recruiter evidence	Test subject/body/thread language with synthetic mail.
Application missed	Weak confirmation evidence	Use testPortalApplicationEmails and inspect detection score.
Status not updated	No reliable row match	Check thread/company/position/recruiter evidence.
Backfill appears stuck	Trigger or lock delay	Run checkAllJobsBackfillProgress and inspect Executions.
Dashboard shows zero	All Jobs not populated yet	Complete live or historical population first.

Gmail Job Tracker - Professional Project Guide

Built for a privacy-safe public GitHub portfolio.